ARSHIA MUBIAS SHAIK

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Master's student in Computer Science at USC specializing in **AI/ML, generative AI, and healthcare fairness.** Skilled in building **production-ready applications**. Experienced in **full-stack development, cloud architecture (AWS, Azure, GCP**), and **deep learning frameworks (PyTorch, TensorFlow, Transformers)**. Expertise **in LLM fine-tuning, and scalable system design**. Led cloud migrations, engineered high-throughput systems, and developed educational content with global reach. **Coursework includes Deep Learning, Natural Language Processing, Privacy-first Machine Learning, and Database Systems.**

EDUCATION

University of Southern California Los Angeles, California
Master of Science, Computer Science (GPA: 3.81) August 2024-Expected Graduation May 2026

Savitribai Phule Pune University Pune, India
Bachelor of Technology, Computer Science (CGPA: 9.41/10) August 2019-May 2023

ACADEMIC PROJECTS

Cognitive Load Reduction Agent | *Python, FunctionGemma 270M, LoRA, MediaPipe, pyttsx3* February 2026 – Present

- Built at the Google DeepMind × InstaLILY Hackathon; fine-tuned FunctionGemma 270M with LoRA on a custom multimodal signal-to-action dataset (webcam, keyboard/mouse, OS state) for real-time cognitive load scoring.
- Implemented an agentic perceive–reason–act–observe–adapt loop with persistent on-device memory, enabling fully private, real-time cognitive load regulation with zero cloud dependency.

Alzi – AI Assistant App for Alzheimer's Patients | *Python, Flutter, LMNT, Flask, Firebase, InsightFace* June 2025 – July 2025

- Engineered a Flutter-based mobile app at UC Berkeley's AI Hackathon integrating a fine-tuned Gemini Pro LLM, Firebase, InsightFace, and Google Calendar API for real-time memory recall and daily routine support for Alzheimer's patients.
- Operationalized an LLM-driven voice assistant with LMNT voice cloning and a full-stack caregiver dashboard with access controls, geofenced alerts, and SOS triggers to enhance caregiver oversight and patient safety.

Music Genre Transfer | *Generative Adversarial Network, Transformer, VAE, MusicGen , Gradio* January 2025 – May 2025

- Implemented and benchmarked multiple deep learning architectures (Transformer, CycleGAN, StarGAN, MusicGen) on the GTZAN dataset for spectrogram-based audio genre transfer while preserving melodic structure.
- Achieved optimal results with prompt-conditioned MusicGen, demonstrating superior genre-accurate transformations compared to traditional generative models.
- Integrated real-time Gradio web interface enabling interactive genre transformation of audio clips with live evaluation.

Bias Detection and Mitigation in Clinical LLMs | *Python, Pandas, Fairlearn, PyTorch, Hugging Face* January 2025 – May 2025

- Fine-tuned BioGPT and ClinicalBERT on 40,000+ MIMIC-III records for mortality prediction, length-of-stay estimation, and treatment recommendations, evaluating fairness across demographic subgroups.
- Applied Fairlearn's ThresholdOptimizer and counterfactual data augmentation to mitigate bias, reducing Equalized Odds disparities by 50% across ethnicity, gender, religion, and marital status while maintaining model accuracy.

EXPERIENCE

Tech Mahindra, SDE Intern | *AWS, Spring Boot, SQL, RESTful APIs, MLOps* July 2022-September 2022

- Engineered full-stack web application using Spring Boot and AWS S3, implementing RESTful APIs with optimized caching that improved data retrieval efficiency by 25%.
- Designed SQL database schema handling 10,000+ daily transactions, implementing indexing and normalization strategies that enhanced query performance by 40%.
- Developed an intelligent data orchestration framework synchronizing real-time writes across AWS S3 and SQL databases, incorporating ML-driven validation to detect anomalies and ensure data integrity during ingestion.

MyLearning Plus Pvt Ltd, Cloud Intern | *AWS, GCP, Python* April 2022-May 2022

- Led the end-to-end migration of the SaaS platform RIZEE from GCP to AWS, leveraging EC2, S3, RDS Multi-AZ, and CloudWatch monitoring with resource optimization strategies that reduced operational costs by 30%.
- Deployed ML inference pipelines on AWS Lambda and SageMaker, reducing latency by 45% for real-time predictions serving 5,000+ daily users.

Defence Research and Development Organisation, Research Intern | *Python, Kalman Filtering* July 2021-September 2021

- Architected radar data fusion pipeline using Python and Kalman filtering to resolve duplicate object detections across overlapping radar coverage, ensuring single-source tracking for surveillance systems.
- Conducted research on AI classification models for identifying airborne object types from radar telemetry data in national defense contexts.

SKILLS

- **ML & AI:** Deep Learning, LLMs, Generative AI, Agentic AI Systems, RAG, Transformers, Prompt Engineering, Claude CLI
- **Frameworks & Libraries**: PyTorch, TensorFlow, Scikit-learn, Keras, LangChain, OpenAI API, OpenCV, Pandas, NumPy
- **App & Web Dev:** Python, Flutter, Dart, JavaScript, HTML/CSS, Unity, C#, .NET, Spring Boot, Flask, RESTful APIs, Firebase
- **Cloud, Data & Platforms:** AWS, GCP, Microsoft Azure, Docker, Kubernetes, CI/CD Pipelines, GitLab, GitHub Actions
- **Databases & Tools**: SQL, MongoDB, Firebase, MySQL, PostgreSQL, Git, Linux, VS Code, Postman, Jupyter, Google Colab
- **Certifications:** AWS (SAA, CCP) | Azure Fundamentals & AI Fundamentals | GCP Associate Cloud Engineer